Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A charged particle with charge q enters a region of constant, uniform and mutually orthogonal field $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{v}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then

(A) $\vec{v}=\frac{\vec{E}\times\vec{B}}{B^2}$

(B) $\vec{v}=\frac{\vec{B}\times\vec{E}}{B^2}$

(C) $\vec{v}=\frac{\vec{E}\times\vec{B}}{E^2}$

(D) $\vec{v}=\frac{\vec{E}\times\vec{B}}{E^2}$

Q.2

A $10\mathrm{eV}$ electron is circulating in a plane at right angles to a uniform field at magnetic induction $10^{-4}\ \mathrm{Wb/m^2}(=1.0$ gauss $)$. The orbital radius of the electron is

(A) 12 cm

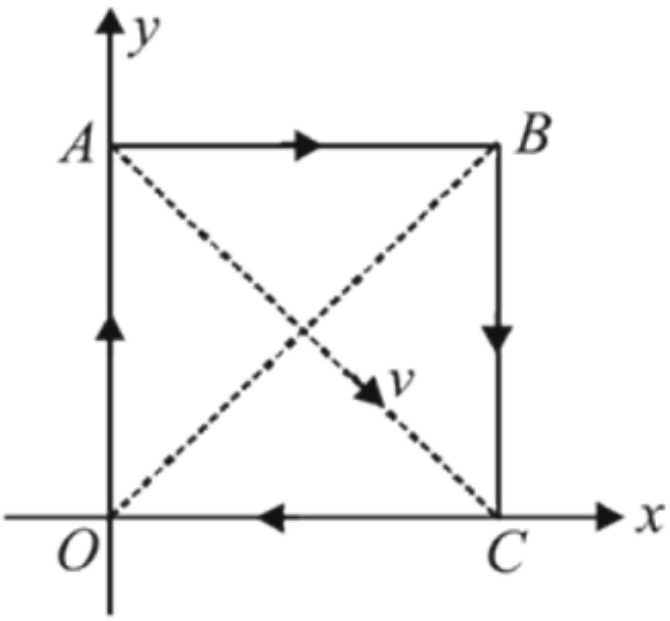
(B) 16 cm

(C) 11 cm

(D) 18 cm

Q.3

$OABC$ is current carrying square loop an electron is projected from the centre of loop along its diagonal AC as shown. Unit vector in the direction of initial acceleration will be



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(A) $\hat{k}$

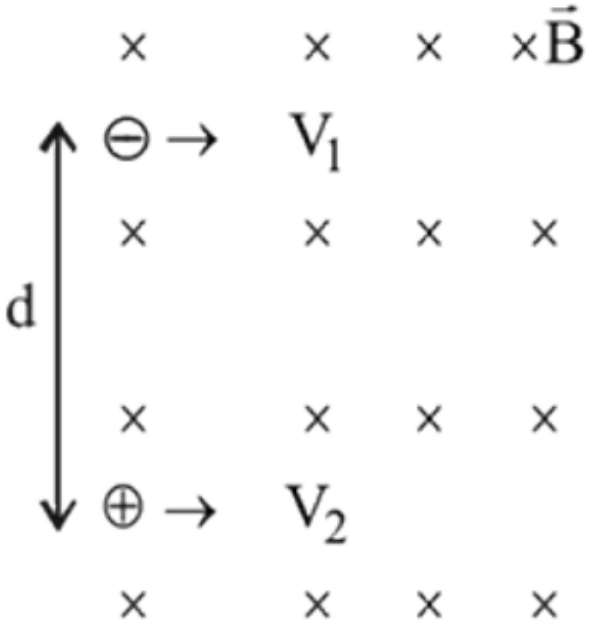
(B) $-\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(C) $-\hat{k}$

(D) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Q.4

Two identical particles having the same mass m and charges $+q$ and $-q$ separated by a distance d enter a uniform magnetic field B directed perpendicular to paper inwards with in speeds v_1 and v_2 as shown in Fig. The particles will not collide if



(A) $d > \frac{m}{Bq}(v_1 + v_2)$

(B) $d < \frac{m}{Bq}(v_1 + v_2)$

(C) $d > \frac{2m}{Bq}(v_1 + v_2)$

(D) $v_1 = v_2$

Q.5

A cyclotron is operated at an oscillator frequency of 24MHz and has a dee radius $R = 60\text{ cm}$. What is magnitude of the magnetic field B (in Tesla) to accelerate deuterons (mass $= 3.34 \times 10^{-27}$) kg?

(A) 9.5

(B) 7.2

(C) 5.0

(D) 3.2

Q.6

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A charged particle of specific charge (charge/ mass) α is released from origin at time $t = 0$ with velocity $\vec{v} = v_0(\hat{i} + \hat{j})$ in uniform magnetic field $\vec{B} = B_0\hat{i}$. Coordinates of the particle at time $= \pi/(B_0\alpha)$ are

(A) $\left(\frac{v_0}{2B_0\alpha}, \frac{\sqrt{2}v_0}{\alpha B_0}, \frac{-v_0}{B_0\alpha}\right)$

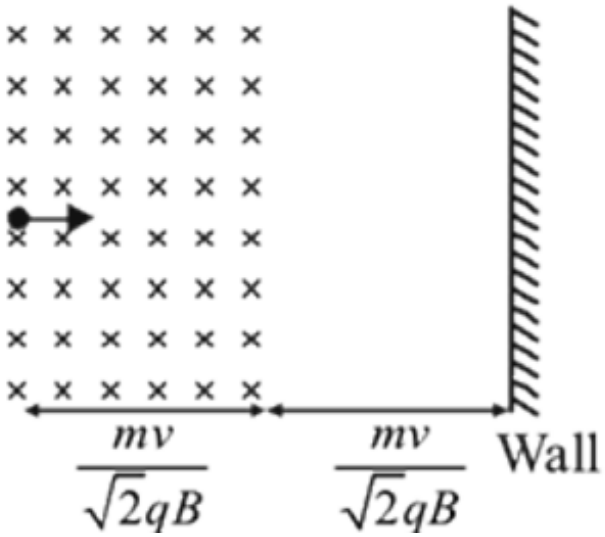
(B) $\left(\frac{-v_0}{2B_0\alpha}, 0, 0\right)$

(C) $\left(0, \frac{2v_0}{B_0\alpha}, \frac{v_0\pi}{2B_0\alpha}\right)$

(D) $\left(\frac{v_0\pi}{B_0\alpha}, 0, \frac{-2v_0}{B_0\alpha}\right)$

Q.7

A particle of mass m and charge q enters a region of magnetic field (as shown) with speed v. There is a region in which the magnetic field is absent, as shown. The particle after entering the region collides elastically with a rigid wall. Time after which the velocity of particle becomes antiparallel to its initial velocity is



(A) $\frac{m}{2qB}(\pi + 4)$

(B) $\frac{m}{qB}(\pi + 2)$

(C) $\frac{m}{4qB}(\pi + 2)$

(D) $\frac{m}{4qB}(2\pi + 3)$

Q.8

A particle of mass m and charge q moves with a constant velocity v along the positive $\bar{x}$ direction. It enters a region containing a uniform magnetic field B directed along the negative z -direction, extending from $x = a$ to $x = b$. The minimum value of v required so that the particle can just enter the region $x > b$ is

(A) qbB/m

(B) $q(b - a)B/m$

(C) qaB/m

(D) $q(b + a)B/2\,m$

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Q.9

Two charged particles having charge Q and $-Q$, and masses m and $4m$ respectively enters in uniform magnetic field B at an angle θ with magnetic field from same point with speed v. The displacement from starting point where they will meet again, is

(A) $\frac{2\pi m}{QB} v \sin\theta$

(B) $\frac{2\pi n}{QB} v \cos\theta$

(C) $\frac{8\pi m}{QB} v \cos\theta$

(D) $\frac{12\pi m}{QB} v \cos\theta$

Q.10

An electron of mass m is accelerated through a potential difference of V and then it enters magnetic field of induction B normal to the lines. Then the radius of the circular path is

(A) $\sqrt{\frac{2eV}{m}}$

(B) $\sqrt{\frac{2Vm}{eB^2}}$

(C) $\sqrt{\frac{2Vm}{eB}}$

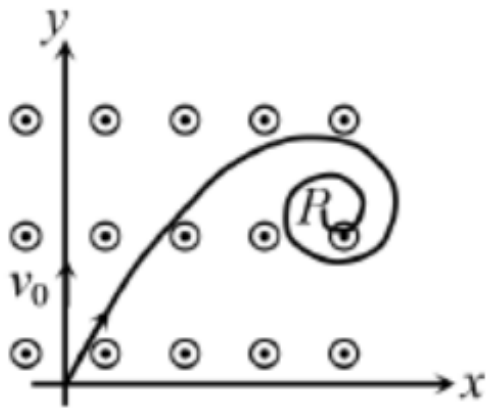
(D) $\sqrt{\frac{2Vm}{e^2B}}$

Q.11

A particle having mass m, charge q is projected with velocity v_0 along y -axis in a region of uniform magnetic field B_0 which is outward and perpendicular to the plane of the paper as shown in the figure. The particle is continuously subjected to a frictional force which varies with velocity as $\vec{F}_r = -\alpha\vec{v}$, where α is a constant. Consequently the particle moves on a spiral path till it comes to rest at point P. The x - co-ordinate of point P is

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(Take $\alpha = 10^{-3}\ \text{kg/s}, \text{q} = 10^{-3}\text{C}, B_0 = 1\ \text{T}, v_0 = 1\ \text{m/s}, \text{m} = 20\text{gm}$)



(A) $10\ \text{m}$

(B) $4\ \text{m}$

(C) $6\ \text{m}$

(D) $8\ \text{m}$

Q.12

A proton of mass $1.67 \times 10^{-27}\ \text{kg}$ and charge $1.6 \times 10^{-19}\text{C}$ is projected with a speed of $2 \times 10^6\ \text{m/s}$ at an angle of 60° to the X - axis. If a uniform magnetic field of 0.104 Tesla is applied along Y - axis, the path of proton is

(A) A circle of radius $= 0.2\ \text{m}$ and time period $\pi \times 10^{-7}\ \text{s}$

(B) A circle of radius $= 0.1\ \text{m}$ and time period $2\pi \times 10^{-7}\ \text{s}$

(C) A helix of radius $= 0.1\ \text{m}$ and time period $2\pi \times 10^{-7}\ \text{s}$

(D) A helix of radius $= 0.2\ \text{m}$ and time period $4\pi \times 10^{-7}\ \text{s}$

Q.13

A particle of charge per unit mass α is released from origin with a velocity $\vec{v} = v_0\hat{i}$ in a uniform magnetic field $\vec{B} = -B_0\hat{k}$. If the particle passes through $(0, y, 0)$, then y is equal to

(A) $-\frac{2v_0}{B_0\alpha}$

(B) $\frac{v_0}{B_0\alpha}$

(C) $\frac{2v_0}{B_0\alpha}$

(D) $-\frac{v_0}{B_0\alpha}$

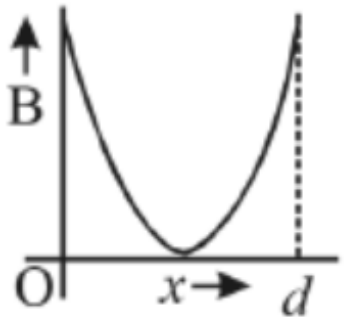
Q.14

Two parallel beams of protons and electrons, carrying equal currents are fixed at a separation d. The protons and electrons move in opposite directions. P is a point on a line

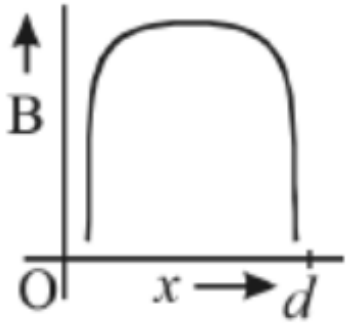
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joining the beams, at a distance x from any one beam. The magnetic field at P is B. If B is plotted against x, which of the following best represents the resulting curve

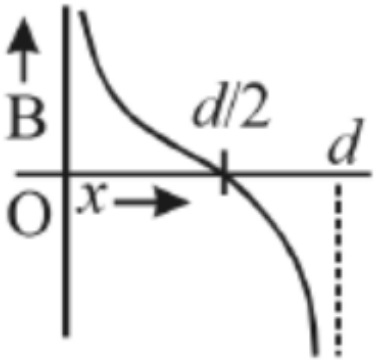
(A)



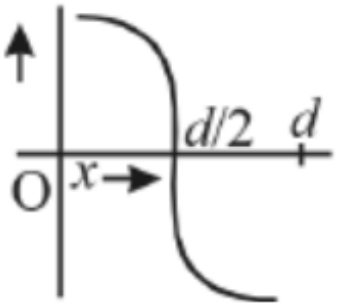
(B)



(C)



(D)



Q.15

A $2\mu C$ charge moving around a circle with a frequency of 6.25×10^{12} Hz produces a magnetic field 6.28 tesla at the centre of the circle. The radius of the circle is

(A) 2.25 m

(B) 0.25 m

(C) 13.0 m

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(D) $1.25\ \mathrm{m}$

Q.16

A current I flows in an infinitely long wire with cross section in the form of a semi-circular ring of radius R. The magnitude of the magnetic induction along its axis is:

(A) $\frac{\mu_0 I}{2\pi^2 R}$

(B) $\frac{\mu_0 I}{2\pi R}$

(C) $\frac{\mu_0 I}{4\pi R}$

(D) $\frac{\mu_0 I}{\pi^2 R}$

Q.17

A particle of mass m and charge q, accelerated by potential difference V enters a region of uniform transverse magnetic field B. If d is the thickness of region of magnetic field, then the deviation of the particle from initial direction when it leaves the field is

(A) $\sin^{-1}\left[\mathrm{Bd}\left(\frac{\mathrm{q}}{2\mathrm{mV}}\right)^{1/2}\right]$

(B) $\cos^{-1}\left[\mathrm{Bd}\left(\frac{\mathrm{q}}{2\mathrm{mV}}\right)^{1/2}\right]$

(C) $\tan^{-1}\left[\mathrm{Bd}\left(\frac{\mathrm{q}}{2\mathrm{mV}}\right)^{1/2}\right]$

(D) zero

Q.18

A particle of mass m and charge q moving with constant velocity v along positive x direction enters a region containing a uniform magnetic fileld B directed along the negative z direction, extending from $x = a$ to $x = b$. For the particle to enter the region $x > b$, the minimum value of v is

(A) $\frac{qbB}{m}$

(B) $\frac{q(b-a)B}{m}$

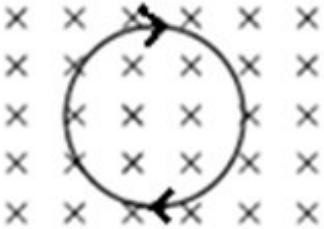
(C) $\frac{qaB}{m}$

(D) $\frac{q(b+a)B}{2m}$

Q.19

A circular coil carrying current I is placed in a region of uniform magnetic field acting perpendicular to the coil as shown in the figure. Mark the correct option.

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(A) coil expands

(B) coil contracts

(C) coil moves left

(D) coil moves right

Q.20

Identify the correct statement among the following:

(A) cyclotron frequency is dependent on speed of the charged particle.

(B) kinetic energy of charged particle in cyclotron does not dependent on its mass.

(C) cyclotron frequency does not depend on the speed of the charged particle.

(D) kinetic energy of charged particle in cyclotron is independent of its charge.

Answers & Solutions

Q.1 Answer:

$\vec{v} = \frac{\vec{E} \times \vec{B}}{B^2}$

Solution:

The forces acting on the particle by magnetic and electric fields are

$\vec{F}_m = q(\vec{v} \times \vec{B})$ **and** $\vec{F}_e = q\vec{E}$

The direction of force $\vec{F}_e$ **is along the direction of** $\vec{E}$ **and the direction of force** $\vec{F}_m$ **is perpendicular to vectors** $\vec{v}$ **and** $\vec{B}$**. Since the velocity of the particle remains unchanged, no net force acts on it. Hence,** $F_m = F_e$ **and the direction of** $\vec{F}_m$ **must be opposite to that of** $\vec{F}_e$**, as shown in the figure. From the lefthand screw rule, it follows that** $\vec{v}$ **should be directed perpendicular to the plane of the page and towards the reader. Thus,**

$$q(\vec{v} \times \vec{B}) = -q\vec{E}$$
$$\vec{v} \times \vec{B} = -\vec{E}$$

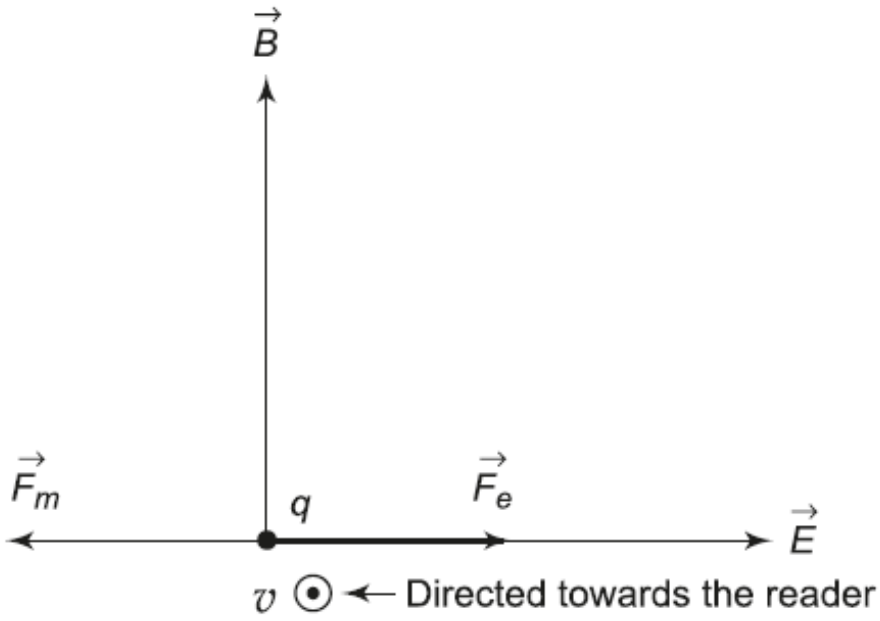
Since $\vec{v}$ **is perpendicular to** $\vec{B}$**, the magnitude of** $\vec{v}$ **is**

$v = \frac{E}{B}$

Hence choices (c) and (d) are wrong because they give $v = \frac{B}{E}$**. Since** $\vec{v}$ **is directed towards the reader, it follows that the direction of vector** $\vec{v}$ **is the same that of vector** $(\vec{E} \times \vec{B})$ **and not of vector** $(\vec{B} \times \vec{E})$

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Hence the correct choice is (b).



Q.2 Answer:

11 cm

Solution:

K.E. of electron $= 10\text{eV} \Rightarrow \frac{1}{2}mv^2 = 10\text{eV}$

$\Rightarrow \frac{1}{2}\left(9.1 \times 10^{-31}\right) v^2 = 10 \times 1.6 \times 10^{-19}$

$\Rightarrow v^2 = \frac{2 \times 10 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}$

$\Rightarrow v^2 = 3.52 \times 10^{12} \Rightarrow v = 1.88 \times 10^6 \text{ m}$

Also, we know that for circular motion $\frac{mv^2}{r} = \text{Bev} \Rightarrow r = \frac{mv}{Be} = 11 \text{ cm}$

Q.3 Answer:

$-\left(\frac{\hat{\text{i}}+\hat{\text{j}}}{\sqrt{2}}\right)$

Solution:

$\vec{B} = -c\hat{k}$, **and** $\vec{v} = v\cos 45^\circ \hat{i} - v\sin 45^\circ \hat{j}$

$= \frac{v}{\sqrt{2}}\hat{i} - \frac{v}{\sqrt{2}}\hat{j}.$

Thus, $\vec{F} = -e(\vec{v} \times \vec{B}) = q\left[\left(\frac{v}{\sqrt{2}}\hat{i} - \frac{v}{\sqrt{2}}\hat{j}\right) \times (-c\hat{k})\right.$

$= ecv\left[\frac{-\hat{i}-\hat{j}}{\sqrt{2}}\right] \quad \therefore \quad \vec{a} = \left[\frac{-\hat{i}-\hat{j}}{\sqrt{2}}\right].$

Q.4 Answer:

$\text{d} > \frac{2\,\text{m}}{\text{Bq}}\left(\text{v}_1 + \text{v}_2\right)$

Solution:

The particle will not collide if $\text{d} > 2\left(\text{r}_1 + \text{r}_2\right)$

or $\text{d} > 2\left(\frac{\text{mv}_1}{\text{Bq}} + \frac{\text{mv}_2}{\text{Bq}}\right)$ **or** $\text{d} > \frac{2\,\text{m}}{\text{Bq}}\left(\text{v}_1 + \text{v}_2\right)$

Q.5 Answer:

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3.2

Solution:

Given: $f = 24 \times 10^6$ Hz, $R = 0.60$ m

We know that, $R = \frac{mv}{qB} \therefore \quad B = \frac{mv}{qR}$

where $v = \omega R = 2\pi f R$

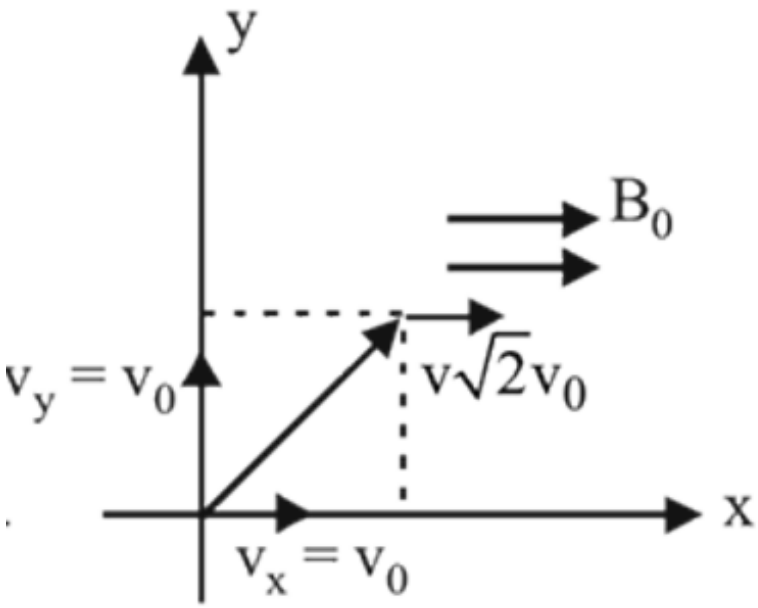
$= 2\pi \times 24 \times 10^6 \times 0.60 = 9.04 \times 10^7$ m/s

From(i), $B = \frac{\left(3.34\times10^{-27}\right)\left(9.04\times10^{7}\right)}{16\times10^{-19}\times060} = 3.2$ T

Q.6 Answer:

$\left(\frac{v_0\pi}{B_0\alpha}, 0, \frac{-2v_0}{B_0\alpha}\right)$

Solution:



the helix

The give time $t = \frac{\pi}{B_0\alpha} = \frac{T}{2}$

$\therefore$ **Coordinates of particle at time** $t = T/2$ **would be** $(vxT/2, 0, -2r)$

Here, $r = \frac{mv_0}{B_0q} = \frac{v_0}{B_0\alpha}$

$\therefore$ **The coordinate are** $\left(\frac{v_0\pi}{B_0\alpha}, 0, \frac{-2v_0}{B_0\alpha}\right)$

Q.7 Answer:

$\frac{m}{2qB}(\pi + 4)$

Solution:

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$r = \frac{mv}{qB}\frac{\frac{mv}{rv}}{r} = \frac{\frac{\sqrt{2}qB}{qB}}{=}\frac{1}{\sqrt{2}}$

or $\theta = \frac{\pi}{4}$

Time to complete the circle $(2\pi), T = \frac{2\pi m}{qB}$

$\therefore$ **time taken to traverses** $\frac{\pi}{4}, t = \frac{\pi m}{4qB}$

Time taken to travel horizontal distance $t_1 = \frac{\frac{mv}{\sqrt{2}qB}}{\frac{v}{\sqrt{2}}} = \frac{m}{qB}$

Total time taken $= 2\mathrm{t} + 2\mathrm{t}_1 = \frac{m}{2qB}(\pi + 4)$

Q.8 Answer:

$q(\mathrm{b} - \mathrm{a})\mathrm{B}/\mathrm{m}$

Solution:

$$\frac{mv^2}{r} = qvB \text{ or } v = \frac{qB}{m}(r) \quad \therefore \quad v_{\min} = \frac{qB}{m}(r_{\min}) = \frac{qB}{m}(b - a)$$

Q.9 Answer:

$\frac{8\pi m}{QB} v \cos\theta$

Solution:

Time period for first $= \frac{2\pi m}{QB}$

Time period for second $= \frac{8\pi m}{QB}$

They will meet again after $\frac{8\pi m}{QB}$

So, displacement in this time interval $= \frac{8\pi m}{QB} v \cos\theta$

$\therefore$ **(c)**

Q.10 Answer:

$\sqrt{\frac{2Vm}{eB^2}}$

Solution:

$Bev = \frac{mv^2}{r}$ **or** $r = \frac{mv}{Be} = \frac{\sqrt{2Km}}{Be}$

As the electron has been accelerated from rest through a potential difference of V **volt,**

then $K = eV$

$\therefore \quad r = \sqrt{\frac{2meV}{B^2e^2}} = \sqrt{\frac{2mV}{B^2e}}$

$\therefore$ **(b)**

Q.11 Answer:

10 m

Solution:

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$q\left[v_x\hat{i}+v_y\hat{j}\right]\times B_0\hat{k}-\alpha\left[v_x\hat{i}+v_y\hat{j}\right]=m\frac{d\vec{v}}{dt}$ ...(i)

$\therefore \quad a_y=-\frac{1}{m}\left[qv_xB_0+\alpha v_y\right]$

and $a_x=\frac{1}{m}\left[qB_0v_y-\alpha v_x\right]$...(ii)

$\therefore$ At $t=0, v_y=v_0$ and $v_x=0$ and $t=t, \quad v_x=0, v_y=0$

$\therefore \quad v_0=\frac{1}{m}\left[qB_0x+\frac{\alpha^2x}{qB_0}\right] \quad \Rightarrow \quad x=\frac{qB_0mv_0}{\alpha^2+q^2B_0^2}=10\text{ m}$

$\therefore$ (a)

Q.12 Answer:

A helix of radius $=0.1$ m and time period $2\pi\times10^{-7}$ s

Solution:

Path of the proton will be a helix of radius $r=\frac{m\sin\theta}{qB}$

(where $\theta=$ angle between $\vec{B}$ and $\vec{v}$)

$\Rightarrow \quad r=\frac{1.67\times10^{-27}\times2\times10^6\times\sin30^\circ}{1.6\times10^{-19}\times0.104}=0.1\text{ m}$

Time period $T=\frac{2\pi m}{qB}=\frac{2\pi\times1.67\times10^{-27}}{1.6\times10^{-9}\times0.104}=2\pi\times10^{-7}\text{sec}$

$\therefore$ (c)

Q.13 Answer:

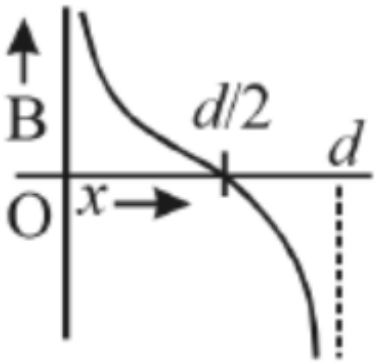
$\frac{2v_0}{B_0\alpha}$

Solution:

$y=2r=\frac{2mv_0}{qB_0}=\frac{2v_0}{B_0\alpha}$

(c)

Q.14 Answer:



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Solution:

The magnetic field at points to the right of the proton beam acts perpendicular to the paper inwards (x). The magnetic field at points to the left of the electron beam acts perpendicular to the paper outwards ())
Magnetic field at mid point M is zero.
Magnetic field at the points closer to proton beam acts perpendicular to the paper inwards (i.e. (x)) and at the points closer to electron beam it acts outwards i.e. ($\cdot$). In the given options graph (c) satisfies all the conditions.

Q.15 Answer:

$1.25 \mathrm{~m}$

Solution:

Magnetic field at the centre of the current loop is

$$\begin{gathered}\mathrm{B}=\frac{\mu_0 2 \pi \mathrm{I}}{4 \pi \mathrm{R}} \\ \text { or, } \mathrm{B}=\frac{\mu_0 2 \pi \mathrm{qv}}{4 \pi \mathrm{R}}, \mathrm{R}=\frac{\mu_0 2 \pi \mathrm{qv}}{4 \pi \mathrm{B}}\end{gathered}$$

Substituting the given values, we get

$$\mathrm{R}=\frac{4 \pi \times 10^{-7} \times 2 \pi \times 2 \times 10^{-6} \times 6.25 \times 10^{12}}{4 \pi \times 6.28}=1.25 \mathrm{~m}$$

Q.16 Answer:

$\frac{\mu_0 I}{\pi^2 R}$

Solution:

Current in a small element, $dI=\frac{d\theta}{\pi} I$ Magnetic field due to the element
$dB=\frac{\mu_0}{4\pi} \frac{2dI}{R}$
The component $dB \cos\theta$, of the field is cancelled by another opposite component. Therefore,

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$B_{net} = \int dB \sin\theta = \frac{\mu_0 I}{2\pi^2 R} \int_0^{\pi} \sin\theta d\theta = \frac{\mu_0 I}{\pi^2 R}$

Q.17 Answer:

$\sin^{-1}\left[\mathrm{Bd}\left(\frac{\mathrm{q}}{2\mathrm{mV}}\right)^{1/2}\right]$

Solution:

$\frac{1}{2}\mathrm{mv}^2 = \mathrm{qV}; \mathrm{v} = \sqrt{\frac{2\mathrm{qV}}{\mathrm{m}}}$

$r = \frac{mv}{qB} = \frac{m}{qB}\sqrt{\frac{2qV}{m}}$

$\mathrm{r} = \frac{1}{\mathrm{B}}\sqrt{\frac{2\mathrm{mV}}{\mathrm{q}}}$

$\mathrm{In}\,\Delta\mathrm{CBD}, \sin\theta = \frac{\mathrm{d}}{\mathrm{r}} = \mathrm{Bd}\sqrt{\frac{\mathrm{q}}{2\mathrm{mV}}}$

$\theta = \sin^{-1}\left\{\mathrm{Bd}\sqrt{\frac{\mathrm{q}}{2\mathrm{mV}}}\right\}$

Q.18 Answer:

$\frac{qb - aB}{m}$

Solution:

$\mathrm{R} = \frac{\mathrm{mv}}{\mathrm{qB}} > \mathrm{b - a}$

$\mathrm{v} = \frac{\mathrm{qB}(\mathrm{b - a})}{\mathrm{m}}$

Q.19 Answer:

coil expands

Solution:

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According to Fleming left hand rule at a small element, it is found that force is acting outward. Hence coil expands as at all points, force is acting outwards.

Q.20 Answer:

cyclotron frequency does not depend on the speed of the charged particle.

Solution:

Cyclotron frequency,

$$n = \frac{1}{T} = \frac{v}{2\pi r}$$

$$n = \frac{v}{2\pi}\frac{Bq}{mv} \qquad \because r = \frac{mv}{Bq}$$

$$n = \frac{Bq}{2\pi m}$$

Thus, cyclotron frequency does not depend upon the speed of the charged particle.

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